ABB Safety Configuration Report

A detailed description of functions and validation procedures can be found in the SafeMove application manual.

1. General Information

Created by:	admin
Creation date:	2026-05-20T17:48:21.0131225+08:00
System name:	15000-101637
Configuration version:	3.00.00
Controller image version:	3.04.00
Checksum:	213B 902E CF8C E749 0D03 494D DDA7 9D2F C430 0FB6 C255 0190 DABE C45C 5116 98E2
Checksum short:	86158EEA

2. Safety Configuration

2.1 Drive Modules

Drive Module 1 Configuration

Max speed manual mode
250.000 mm/s

Drive Module 1 Configuration - ROB_1

Safe brake ramp start speed offset	Elbow offset	Baseframe	
100.000 mm/s	x: -50.000 mm y: -50.000 mm z: 50.000 mm	Position x: 0.000 mm y: 0.000 mm z: 0.000 mm Orientation x: 0.000 deg y: 0.000 deg z: 0.000 deg	

ROB_1 - Upper Arm Geometries

Upper Arm Geometries - Robot_Capsule_1

Type	Radius	Start	End
Capsule	160.000 mm	x: -30.356 mm y: -22.120 mm z: 30.485 mm	x: 186.565 mm y: 0.000 mm z: 100.000 mm

Upper Arm Geometries - Robot_Capsule_2

Type	Radius	Start	End
Capsule	140.000 mm	x: 380.000 mm y: 30.000 mm z: 150.000 mm	x: 520.000 mm y: 10.000 mm z: 150.000 mm

ROB_1 Verified: ______________________________

Drive Module 1 Configuration - Synchronization

Activation	Synchronization status
Software synchronization	No signal

Synchronization - Sync position

Joint	position
1	0.000 deg
2	0.000 deg
3	0.000 deg
4	0.000 deg
5	0.000 deg
6	0.000 deg

Synchronization Verified: ______________________________

Drive Module 1 Configuration - Cyclic Brake Check

CBC: Inactivated

Drive Module 1 Configuration - Tools

Tools - Tool

Activation	Active status	TCP	Orientation	Rigid Body
Permanently active	No signal	x: 0.000 mm y: 0.000 mm z: 0.000 mm	x: 0.000 deg y: 0.000 deg z: 0.000 deg	Mass: 0.000 Kg x: 0.000 mm y: 0.000 mm z: 0.000 mm ix: 0.000 Kgm² iy: 0.000 Kgm² iz: 0.000 Kgm²

Tool - Speed Supervision Points (Flange Coordinates)

Number	X	Y	Z

Tool - Tool Geometries

Tool Geometries - Tool_Capsule

Type	Radius	Start	End
Capsule	75.000 mm	x: 0.000 mm y: 0.000 mm z: 0.000 mm	x: 0.000 mm y: 0.000 mm z: 250.000 mm

Tool Geometries Verified: ____

Tool Verified: ____

Drive Module 1 Configuration - Safe Zones

Safe Zones - Quasi_Static_Contact_Zone_1

Tool speed supervision priority
Base

Quasi_Static_Contact_Zone_1 - Coordinates

Top	Bottom
50.000 mm	-100.000 mm

Vertices

Number	X	Y
1	-1600.000 mm	-1600.000 mm
2	1600.000 mm	-1600.000 mm
3	1600.000 mm	1600.000 mm
4	-1600.000 mm	1600.000 mm

Quasi_Static_Contact_Zone_1 - Floor plan

Quasi_Static_Contact_Zone_1 - Tool Speed Supervisions

Tool Speed Supervisions - Quasi_Static_Contact_TSS

Activation	Function active status	Violation stop category	Violation signal
Permanently active	No signal	Category1Stop	No signal

Max speed
180.000 mm/s

Quasi_Static_Contact_Zone_1 - Tool Force Supervisions

Tool Force Supervisions - Quasi_Static_Contact_TFS

Activation	Function active status	Violation stop category	Violation signal
Permanently active	No signal	Category1Stop	No signal

Max force
70 N

Quasi_Static_Contact_Zone_1 Verified: ____

Safe Zones - Quasi_Static_Contact_Zone_2

Tool speed supervision priority
Base

Quasi_Static_Contact_Zone_2 - Coordinates

Top	Bottom
350.000 mm	0.000 mm

Vertices

Number	X	Y
1	-100.000 mm	-200.000 mm
2	100.000 mm	-200.000 mm
3	100.000 mm	150.000 mm
4	-100.000 mm	150.000 mm

Quasi_Static_Contact_Zone_2 - Floor plan

Quasi_Static_Contact_Zone_2 - Tool Speed Supervisions

Tool Speed Supervisions - Quasi_Static_Contact_TSS_1

Activation	Function active status	Violation stop category	Violation signal
Permanently active	No signal	Category1Stop	No signal

Max speed
180.000 mm/s

Quasi_Static_Contact_Zone_2 - Tool Force Supervisions

Tool Force Supervisions - Quasi_Static_Contact_TFS_1

Activation	Function active status	Violation stop category	Violation signal
Permanently active	No signal	Category1Stop	No signal

Max force
70 N

Quasi_Static_Contact_Zone_2 Verified: ____

Safe Zones - Transient_Contact_Zone

Tool speed supervision priority
Base

Transient_Contact_Zone - Coordinates

Top	Bottom
2000.000 mm	50.000 mm

Vertices

Number	X	Y
1	-1500.000 mm	-1500.000 mm
2	1500.000 mm	-1500.000 mm
3	1500.000 mm	1500.000 mm
4	-1500.000 mm	1500.000 mm

Transient_Contact_Zone - Floor plan

Transient_Contact_Zone - Tool Speed Supervisions

Tool Speed Supervisions - Transient_Contact_TSS

Activation	Function active status	Violation stop category	Violation signal
Permanently active	No signal	Category1Stop	No signal

Max speed
434.000 mm/s

Transient_Contact_Zone Verified: ____

2.2 Stop Configurations

ExternalEmergencyStop

Stop category	Decoupled	
Category1Stop	Coupled	

ExternalEmergencyStop Verified: ____

LocalEmergencyStop

Stop category	
Category1Stop	

LocalEmergencyStop Verified: ____

ProtectiveStop

Mode	Stop category	
Auto	Category1Stop	

ProtectiveStop Verified: ____

3. Safe I/O Configuration

3.1 Global Signals

Name	Type	Default	Offset	Width	Direction	Protected
AutomaticMode	BOOL	0	0	1	output	false
DriveEnable	BOOL	0	1	1	output	false
DriveEnableFeedback	BOOL	0	2	1	output	false
EmergencyStopActivated	BOOL	0	3	1	output	false
EnableSwitch	BOOL	0	4	1	output	false
ExternalEmergencyStopStatus	BOOL	0	5	1	output	false
LocalEmergencyStopStatus	BOOL	0	6	1	output	false
ManualFullSpeedMode	BOOL	0	7	1	output	false
ManualMode	BOOL	0	8	1	output	false
ProtectiveStop	BOOL	0	9	1	output	false
SafetyEnable	BOOL	1	10	1	output	false

3.2 Networks

Feedback

Feedback - Devices

SC_Feedback_Dev

Devices - Signals

Devices - Output

Name	Type	Default	Offset	Width	Direction	Protected
AutomaticMode	BOOL	0	0	1	output	false
DriveEnable	BOOL	0	1	1	output	false
DriveEnableFeedback	BOOL	0	2	1	output	false
EmergencyStopActivated	BOOL	0	3	1	output	false
EnableSwitch	BOOL	0	4	1	output	false
ExternalEmergencyStopStatus	BOOL	0	5	1	output	false
LocalEmergencyStopStatus	BOOL	0	6	1	output	false
ManualFullSpeedMode	BOOL	0	7	1	output	false
ManualMode	BOOL	0	8	1	output	false
ProtectiveStop	BOOL	0	9	1	output	false
SafetyEnable	BOOL	1	10	1	output	false

3.3 Function Mappings

Function	Signal	Mandatory	Description	Group
AutomaticMode	AutomaticMode	true		
DriveEnable	DriveEnable	true		
DriveEnableFeedback	DriveEnableFeedback	true		
EmergencyStopActivated	EmergencyStopActivated	true		
EnableSwitch	EnableSwitch	true		
ExternalEmergencyStopStatus	ExternalEmergencyStopStatus	true		
LocalEmergencyStopStatus	LocalEmergencyStopStatus	true		

3. Safe I/O Configuration

Function	Signal	Mandatory	Description	Group
ManualFullSpeedMode	ManualFullSpeedMode	true		
ManualMode	ManualMode	true		
ProtectiveStop	ProtectiveStop	true		
SafetyEnable	SafetyEnable	true		

4. Combinatorial Logic Configuration

4.1 Pre Logic

Name	Expression

4.2 Post Logic

Name	Expression

Complete functionality verified and tested

Date

Signature